

Exercise 4

Part I: Business Case Instructions

Section	Description
Market Situation	- Define the market situation - Provide supporting information
Opportunity	- Summarize the market opportunity - Describe how the company can benefit
Envisioned Solution	- Describe the envisioned product solution - Summarize how the team arrived at the solution - Describe why this solution is better than alternatives
Objectives and Assumptions	- Describe the market objectives that are being pursued - Define the business assumptions
Costs	- Define product solution costs - List required resources
Strategic Alignment	- Illustrate how this product solution aligns with organizational needs - Summarize how the product concept aligns with organizational goals
Timing	- Describe project timing - Map timing to the market need - Recommend an investment strategy
Cost Benefit Analysis	- Provide a financial analysis—Illustrate costs vs. benefits - Define what happens if the company does not proceed
Risk	- Identify major risks - Document contingency responses/mitigations
Initial Investment and Recommendation	- Define the initial investment requirements - Define long-term investment requirements

Part II: Business Case Blank Template

Section	Description
Market Situation	The global air purifier market is valued at \$6.3 billion (2022) and projected to reach \$13.4 billion by 2030, with a CAGR of 11.4%. Increasing consumer awareness of air quality, pollution, and health risks fuels demand for advanced, ozone-free, low-maintenance purifiers. PP LLC has over ten years in the industry and a strong reputation, giving it a competitive advantage in entering this expanding segment.
Opportunity	Customers are dissatisfied with traditional HEPA purifiers that fail to remove VOCs, viruses, and odors, while also producing ozone. By addressing these shortcomings, PP LLC can expand its customer base by 10% and increase revenue by 25%. This initiative aligns with growing health and environmental awareness trends, giving PP LLC a first-mover advantage in safe, multi-functional purification technology.
Envisioned Solution	Develop and launch a new HEPA air purifier line with three models (Basic, Better, Ultimate) offering: dual filtration, odor elimination, ozone-free operation, reduced noise, and smart maintenance alerts. The team selected this approach after consumer feedback and competitive analysis confirmed demand for an eco-friendly, quiet, and long-lasting purifier. This solution outperforms alternatives by addressing all major consumer pain points simultaneously.
Objectives and Assumptions	Objectives: - Develop three prototype models within six months. - Launch by June to capture summer demand. - Achieve ROI $\geq$ 150% and IRR > 200%. Assumptions: - Stable supply chain and component availability. - Engineering milestones are completed on time. - Marketing campaign increases brand visibility by 20%.
Costs	Initial prototype and design cost: \$80,000. Annual operating and marketing budget: \$30,000. Resources include engineering labor, test equipment, software licenses, and marketing analytics. Expected return: \$200,000/year for two years.
Strategic Alignment	This initiative aligns with PP LLC's corporate goal of expanding market share through product innovation. The new line complements existing HEPA offerings, strengthens the company's sustainability profile, and enhances customer trust by addressing safety and performance gaps.

Timing	Project start: January 5. Design phases: January–February. Development: March–May. Beta testing: May. Market launch: June (ahead of peak summer demand and air pollution season).
Cost Benefit Analysis	Costs: \$110,000 (total). Benefits: Expected revenue of \$400,000 over two years. NPV (10%) = positive; IRR > 200%. Payback period: less than 6 months. If the project is not pursued, PP LLC risks losing market share and brand relevance in a rapidly growing segment.
Risk	Key Risks: Engineering delays, cost overruns, supply chain disruptions, and competitor fast-follow launches. Mitigations: Dedicated project manager for risk oversight, backup suppliers, and incremental development reviews every two weeks.
Initial Investment and Recommendation	Initial Investment: \$80,000 for design and prototype development, with an additional \$30,000 for marketing and resource support during the Plan phase. Recommendation: Proceed with product development and market launch. Financial forecasts, customer demand, and strategic fit strongly justify moving forward.